PaN-Finder Deliverable 4.b:

PaN-Finder Public Documentation

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1 Document Overview

This document provides comprehensive public documentation for the PaN-Finder (Photon and Neutron Federated Knowledge Finder) system. It serves as a complete guide for users at all levels of technical expertise, from citizen scientists exploring open data to domain experts conducting advanced research.

1.1 Purpose

Enable users to effectively search, discover, and utilize scientific data from photon and neutron research facilities across Europe.

1.2 Scope

Covers the public web interface, API usage, practical examples for various user scenarios, technical architecture, and detailed reference information.

2 Audience

This documentation is designed for three primary user groups:

2.1 Non-Technical Users

- Individuals without technical knowledge of system architecture
- Researchers from other disciplines looking for relevant data
- General users interested in scientific data discovery

2.2 Scientific Experts

- Neutron and photon researchers
- Scientists working at research facilities (ESS, ESRF, ILL, MAX IV, PSI, etc.)
- Data analysts and computational scientists
- Experts in materials science, chemistry, physics, and related fields

2.3 Citizen Scientists

- Open data enthusiasts
- Educators and students
- Science communicators
- Public interested in scientific research

3 Introduction to PaN-Finder

3.1 What is PaN-Finder?

PaN-Finder (Photon and Neutron Federated Knowledge Finder) is an **open, AI-powered search tool** designed to help researchers, scientists, and the public discover **datasets, publications, and experimental results** from **neutron and photon research facilities** across Europe. It leverages **natural language processing (NLP)** and **semantic search** to make it easier to find relevant scientific data, even for users without technical expertise.

PaN-Finder is part of the **PaNOSC** (Photon and Neutron Open Science Cloud) initiative, which aims to make scientific data from large-scale research infrastructures **findable, accessible, interoperable, and reusable (FAIR)**.

PaN-Finder is an independent project developed by the European Spallation Source and funded through OSCARS (Open Science Cloud for Accelerator-based Research Services). OSCARS has received funding from the European Commission's Horizon Europe Research and Innovation programme (Grant Agreement No. 101129751). It was developed within the broader PaNOSC ecosystem and has been adopted as a PaNOSC tool for the PaNOSC EOSC node.
The public web portal is accessible at https://pan-finder.panosc.ess.eu/, the project deliverables are available at https://github.com/PaN-Finder/deliverables, while the source code at https://github.com/PaN-Finder/pan-finder

3.2 Data Sources

PaN-Finder currently indexes open data from the following **PaNOSC partner facilities and repositories**:

Facility	Location	Type	Website
ESRF	Grenoble, France	Synchrotron (Photon)	www.esrf.eu
ESS	Lund, Sweden	Spallation (Neutron)	europeanspallationsource.se
ILL	Grenoble, France	Reactor (Neutron)	www.ill.eu
MAX IV	Lund, Sweden	Synchrotron (Photon)	www.maxiv.lu.se
PSI	Villigen, Switzerland	Synchrotron, spallation neutron source, muon source, and X-ray free-electron laser	www.psi.ch
DESY	Hamburg, Germany	Synchrotron (Photon)	www.desy.de
Human Organ Atlas	Grenoble, France	Data Repository	human-organ-atlas.esrf.fr

3.3 Use Cases

For All Users:

- Discover scientific datasets from European research facilities
- Find publications related to specific experiments
- Explore open data from photon and neutron sources

For Scientific Experts:

- Locate open access datasets for analysis and reproducibility studies
- Find experiments with specific parameters (temperature, pressure, energy, etc.)
- Discover related research across different facilities
- Validate and cross-reference experimental results
- Discover additional resources for proposal writing

For Citizen Scientists:

- Access open scientific data for educational purposes
- Learn about photon and neutron research methodologies
- Explore the scientific output of photon and neutron European research infrastructures

4 Getting Started

4.1 Accessing PaN-Finder

The primary way to use PaN-Finder is through the public web portal available at the following URL:

https://pan-finder.panosc.ess.eu/

Any modern web browser, such as Chrome, Firefox, and Safari, should be able to load the application. Make sure that JavaScript is enabled

No installation is required, as the PaN-Finder is a web-based application that runs completely in the browser.

For programmatic usage please use the REST API, see API Documentation.

4.2 User Interface Overview

4.2.1 New search

The PaN-Finder web interface is designed to be **intuitive and user-friendly**. Upon the first visit

or a new search request, the user is presented with the page shown in figure 1.

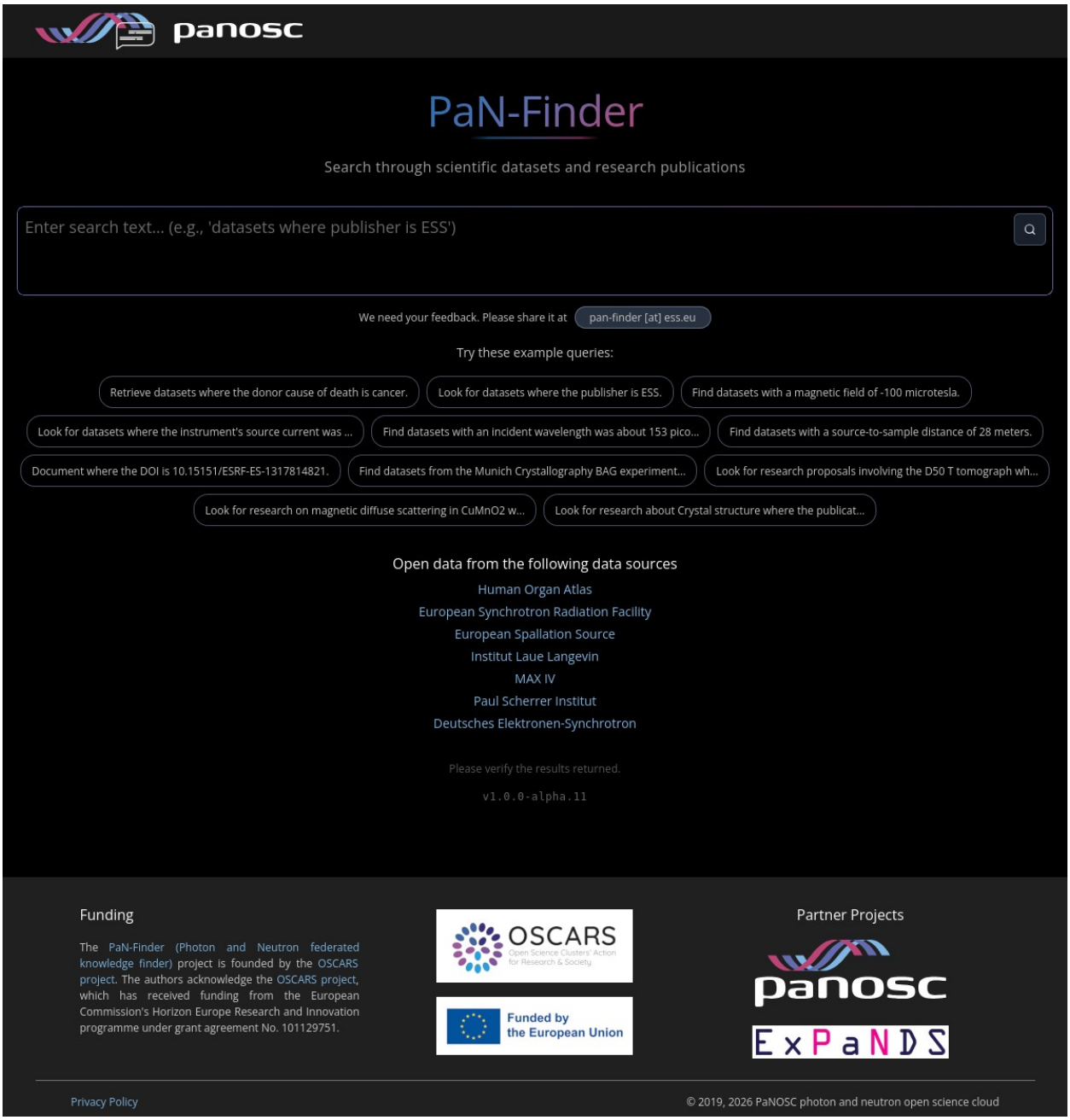


Figure 1. PaN-finder new search

At the top of the page, right below the header, we find the search field. Here the user can enter their query in a natural language, for example "*Show me neutron scattering data for proteins at 100 K*". In order to perform the search, it is necessary to press **Enter** or click the **Search** button. Moving toward the bottom part of the page, the page displays the email address that can be used for support or feedback. Below this, the page presents a set of example queries, such as "*Open data from Human Organ Atlas*", "*Find datasets on graphene*", or "*Show me experiments with temperature between 50 K and 150 K*". These examples are provided to help users formulate effective queries.

4.2.2 Query results

When the user submits a search, the page is continuously refreshed in order to show the progress of the search and which step is currently being performed. Once the results are ready for the user, a final refresh is performed and the results are presented. The final layout of the page is visible in figure 2.

The top of the page consists of the header of the application together with the search box, containing the user query, and the main email. Below, the user finds the results section, which can consist of three independent sections containing up to 20 results in total: matching results, relevant results, and suggested results. This hard limit is set in the backend and can be changed in the backend configuration.
Matching results are results that have a relevance higher than 95%. Relevant results contains results that are predicted to provide relevant information to the user but do not match exactly all the conditions from the query. Suggested results are entries that contain some information that can be useful for the user, but we cannot guarantee they contain relevant information in the context of the user query.
Not all three sections are present in every search. They are presented as needed, based on the relevance scores assigned to each returned result. Each section is styled as a table with one row per result. Each row contains the expansion icon, the DOI of the document, the title, the facility and the relevance score as a percentage. Users can view more information regarding each entry by expanding the row of interest. Expanding and collapsing is achieved by clicking on the expansion icon presented on the left of the row. When the entry is expanded, the user will be able to see an AI-generated relevance explanation, which is created on the fly upon the first expansion. Next, aligned to the right the entry feedback if found, more in the next section, following the user will see the full title, the list of authors, the full abstract, the facility's full name, the instrument used to acquire the associated data, if available, and the date when the data is or will become publicly accessible.
The DOI presented for each entry is a link to the original entry page hosted at the source facility. On this page, users should visit the original page to find additional information and instructions on how to access the associated data.

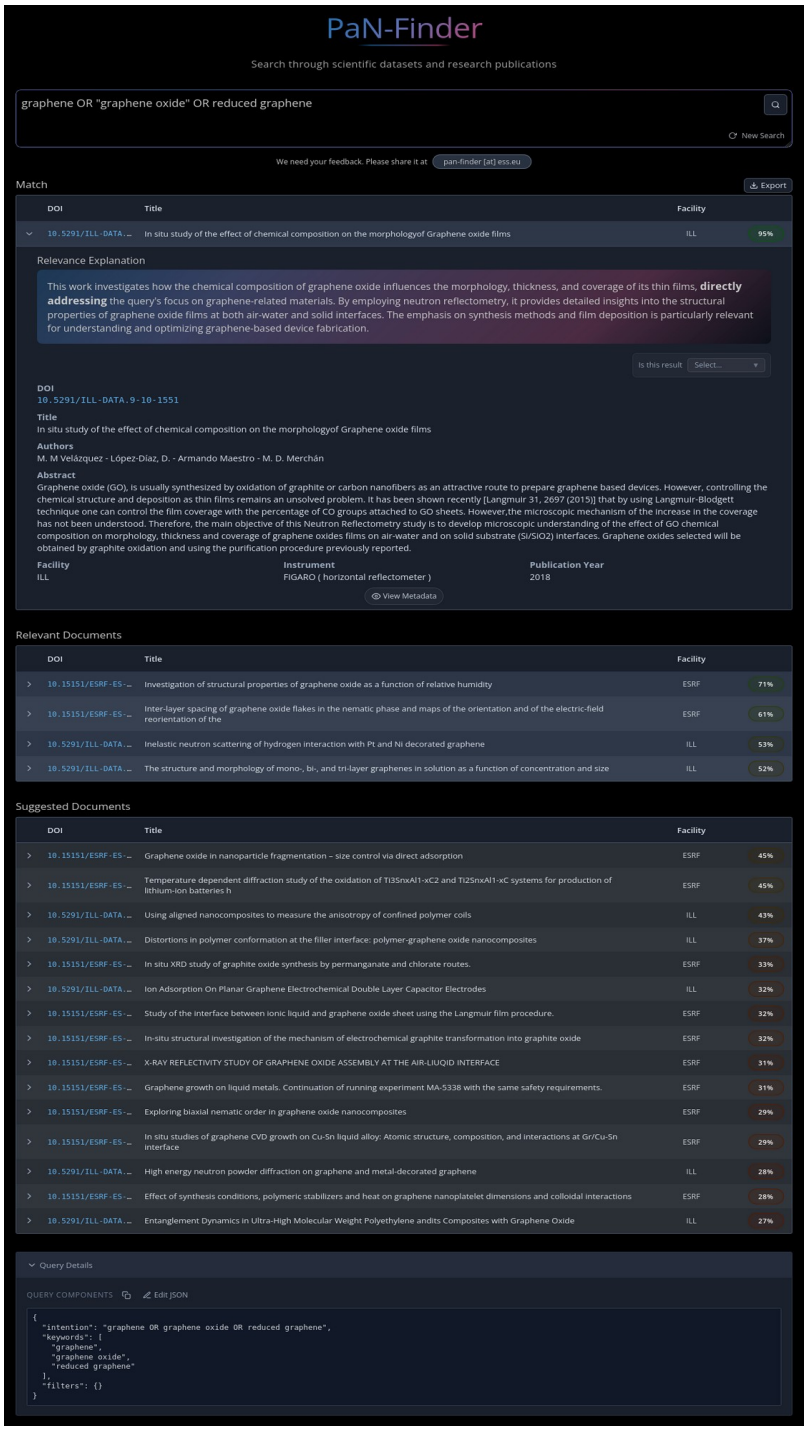


Figure 2. Results screen

In addition to the curated list of information just described, the user can view the raw metadata that our system was able to collect from the facility which has been used during ingestion to create the corresponding entry in the PaN-Finder. The raw metadata can be extensive, the application shows only an excerpt, although a download button allows users to save the full raw metadata locally for further inspection.
Moving towards the bottom of the page, the user will find the query details. This is an advanced feature that allows the user to edit each component that has been extracted from their prompt. The feature requires a deeper knowledge of the PaN-Finder technology, which can be found in all the other deliverables.
In order to use this feature, the user should click on “edit JSON”. The text box containing the query components in JSON format will become editable. Once the user has edited the text, they can submit the new query, skipping the components extraction of the prompt and submitting the customized components directly to the query building stage.

4.2.3 Entry feedback

In order to improve the quality of the service provided, we added a feedback widget to each entry. We rely on our users, more so on domain experts, to collect feedback on each result that we present and their ranking.
Users can select if the specific entry is a **Match**, **Relevant**, **Suggested** or **Not Relevant** at all. Such feedback is collected and will be used in future iterations to improve the scoring engine and expand specific science domain knowledge.

5 User interface details

In this section, we will review in detail all the components of the user interface across the full frontend.

5.1 First search

The first-search screen is shown to the user when the PaN-Finder application is first loaded in the browser, or when the user resets the application after a search.

The first-search screen has five main sections:

- Header
- User prompt
- Example queries
- Facility list or data sources
- Footer

They will be discussed in detail in the following sections

5.1.1 Header

It contains the PaN-Finder logo and the slogan of the project (Figure 3). In future releases more features might be added here.

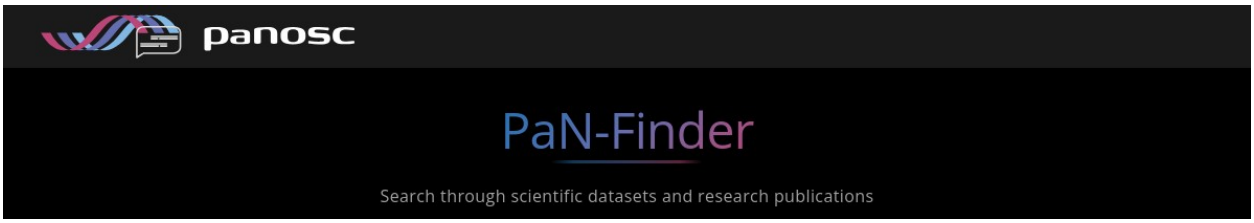


Figure 3. PaN-Finder header

5.1.2 User Prompt

This section presents a prompt box where the user may type in their query (Figure 4). The query can be submitted to the PaN-Finder by pressing the Enter key on the keyboard or by clicking on the magnifying glass icon shown in the top right corner of the prompt box.
Please see the examples section for a range of examples collected from different scientific fields and from different users.

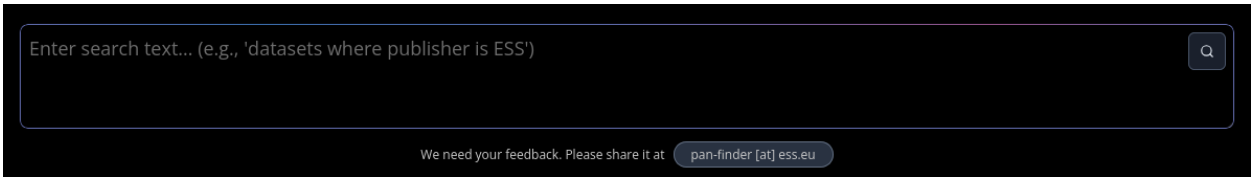


Figure 4. User prompt

Right under the user prompt we added a banner to encourage users to provide feedback which also includes the project's general email address: pan-finder@ess.eu.

5.1.3 Example Queries

This section presents a list of example queries that are offered to the users to discover the functionalities of the PaN-Finder and help users develop a better language to write effective queries (Figure 5).

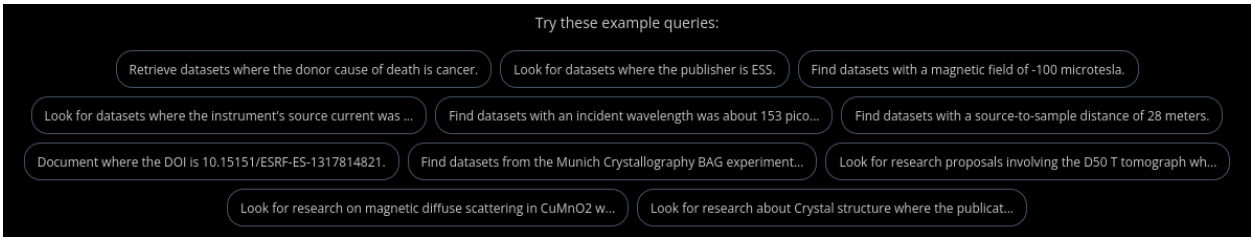


Figure 5. Example queries

The queries included in this section are samples of all the queries that have been used during testing and collected during user interviews.

5.1.4 Facilities and Repositories List

This section, visible in figure 6, lists all the facilities and repositories from which PaN-Finder collects open data and can provide results. It will be updated when new facilities will be added as data sources and their open data collected in the PaN-Finder knowledge base.
This section is also present on the results screen.

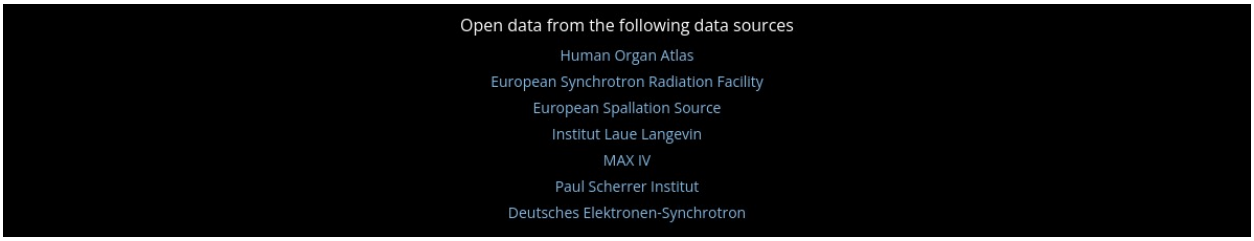


Figure 6. Facilities list

5.1.5 Footer

This section shows the warning to verify the results, the version of the PaN-Finder and provides additional information about the fundings that supported the development of the PaN-Finder and the PaN-Finder community (Figure 7). This section is also present in the results screen.

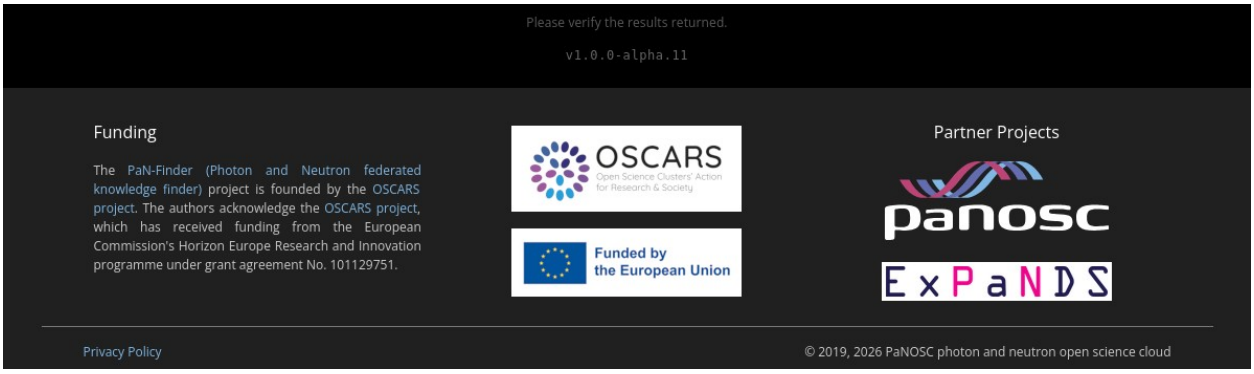


Figure 7. PaN-Finder Footer

5.2 Results screen

As already presented in the previous section, the results page presents the results of the search to the user, together with the prompt entered by the user and the advanced feature that allows expert users to edit the individual components extracted from the query.
The sections presented on the results page are:

- Header
- User prompt
- Match results
- Relevant results
- Suggested results
- Query components
- Facilities and repositories
- Footer

We will not cover the header, facilities and repositories, or the footer as they have already been covered in the previous section and their content does not change.

5.2.1 User prompt

The user-prompt section (Figure 8) presents the query submitted by the user which is entered in the prompt box. It also provides two buttons: the magnifying glass icon and the “new search”. The magnifying-glass icon allows the user to refine the previous query and submit it. The “new search” button with the reload icon clears the query, empties the prompt box and removes the results. It takes the user back to the first screen.

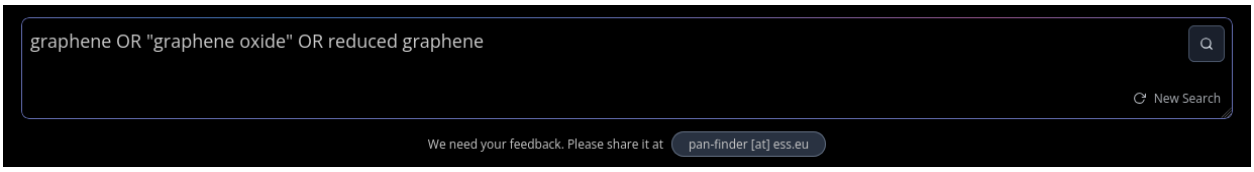


Figure 8. Prompt box with user query

5.2.2 Result categories

The results are presented in three different categories: Match, Relevant and Suggested. A fourth category is listed in the user feedback: non relevant. They are never presented to the user.

Category	**Meaning**	**Relevance**
Match	Results that closely satisfy the query conditions	Above 95%
Relevant	Results related to the query but not satisfying all conditions	Between 95% and the relevant-result threshold determined by ranking/knee-point logic
Suggested	Results that may contain useful related information	Below the relevant-result threshold
Not Relevant	User feedback only indicating that a result should not be presented	

5.2.3 Match results

This section contains the results with computed relevance above 95% (Figure 9). These results are presented as the results estimated to be the entries that the user was exactly looking for. If there are no results with relevance in this range, this section is not shown.
How the results are presented will be discussed in the section “Individual entry information” as all the results sections use the same philosophy and design.
As this is the first results section shown, it also includes the export functionality, visible in the top right corner. The export allows the user to export the results set with all the information presented as a CSV file and enables the use of this information in further processing that the user might want to perform on them.
It is worth noting that the badge reporting the relevance is green. The color transitions from red at zero relevance to green at 100% relevance.

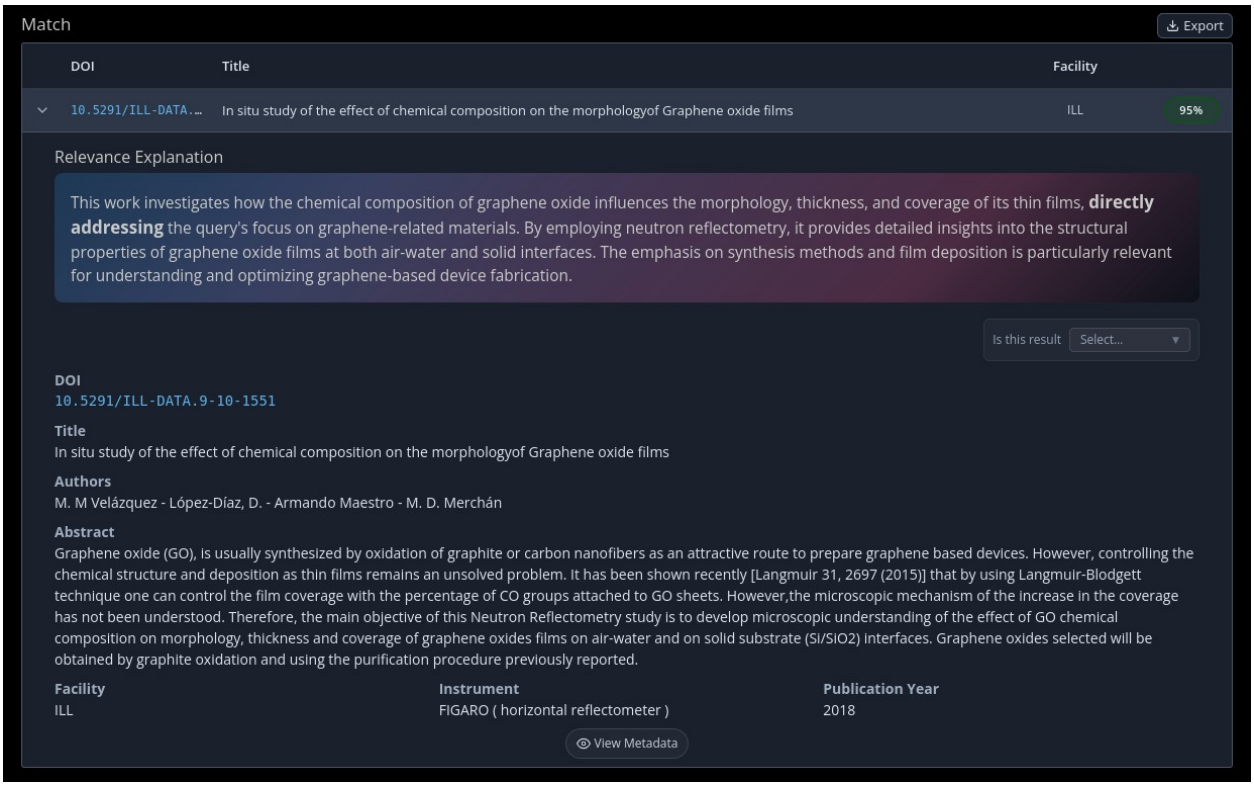


Figure 9. Match results

5.2.4 Relevant results

This section shows results that are relevant to the user but not exactly what they are looking for. Their relevance is at or below 95% but above the threshold selected by the knee-point algorithm. The knee-point algorithm determines the division between Relevant and Suggested results. More information about the knee point algorithm can be found in Deliverable 3. The application is designed not to show this section if the results have low relevance and the knee-point algorithm is not able to define a knee point. This section is visible in figure 10

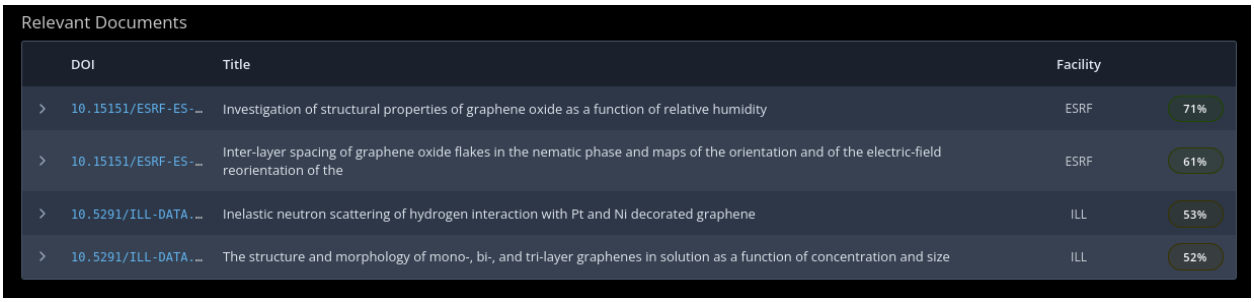


Figure 10. Suggested results

5.2.5 Suggested results

This section shows the results that are somehow connected to the query that the user entered, but are not scoring high relevance (Figure 11). They might contain useful information as they match only some of the conditions expressed in the user query, but they are not necessarily relevant to the user's needs. Their relevance is at or below the threshold found with the knee-point algorithm, until the total number of results reaches 20. As with the other sections, it is shown if there are results that fall in this group. If no results have low relevance, this section is hidden.

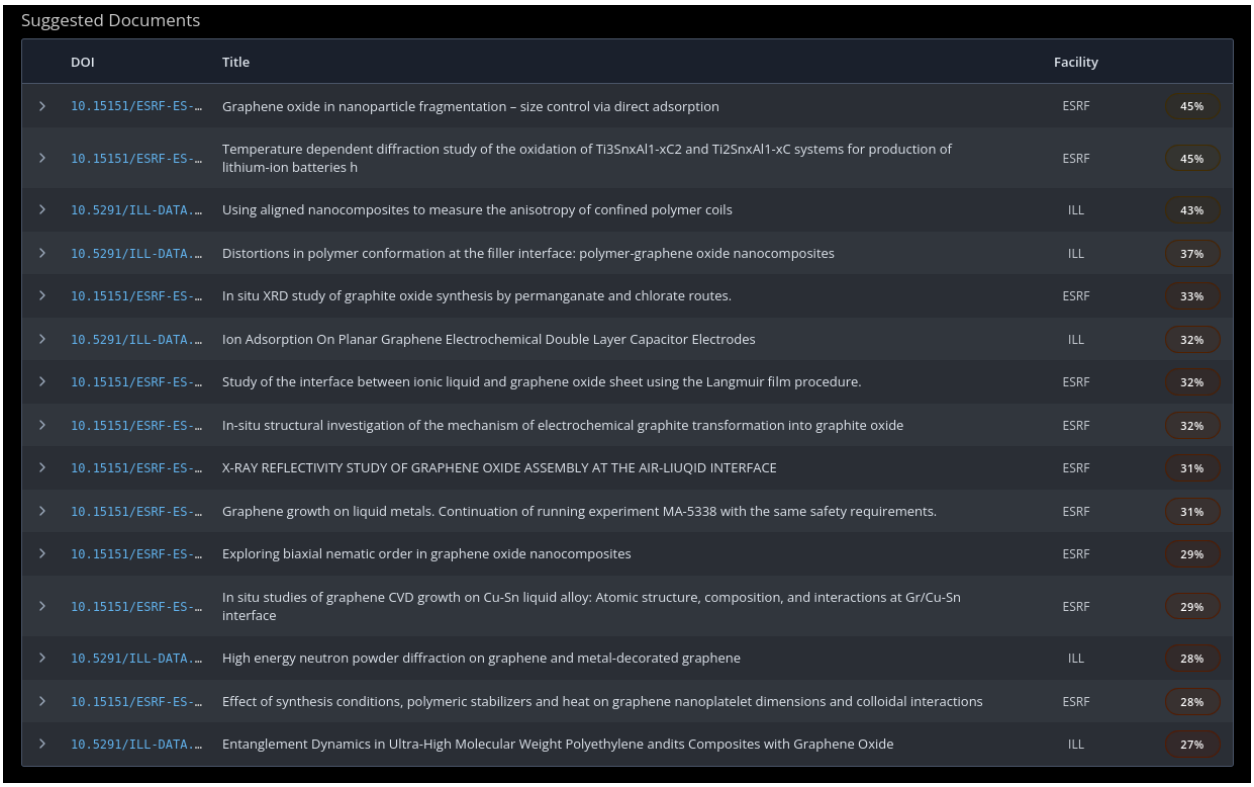


Figure 11. Suggested results

5.2.6 Query components

This section of the UI is mainly directed to advanced users who have domain knowledge and/or technical knowledge of the PaN-Finder (Figure 12). As illustrated in previous deliverables, the PaN-Finder takes the user prompt and tries to extract from it three main components:

- intention
- keywords
- filters

These components are then used to build the database query that retrieves the result set for the user. The PaN-Finder uses an agent to perform the components extraction, therefore the process, although powerful, is non-deterministic, and the results vary and might not be correct.

This section allows the user to inspect the components, which are the results of the first step in the search process. If the user is not satisfied with the component extraction step, they can click the “edit JSON” and edit them directly in JSON format. Once they are satisfied with their

editing, they can submit the new components directly to the engine skipping the extraction step.

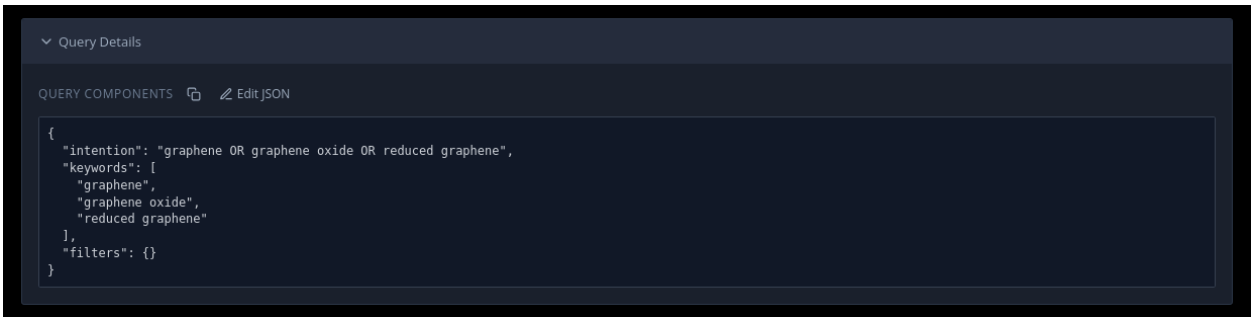


Figure 12. Query components

5.2.7 Individual entry information

Each row in the three results sections has two states: collapsed and expanded. When the results page is presented, all the results are collapsed except for the first and most relevant one, which is expanded automatically. The two states are shown and explained in the next two sections.

5.2.7.1 Collapsed result entry

An entry in a collapsed state occupies a single row of the results section and presents the following information:

- doi
- title
- facility
- color coded score badge

An example can be seen in figure 13.
The user can reach the landing page of the data at the facility by clicking directly the DOI.

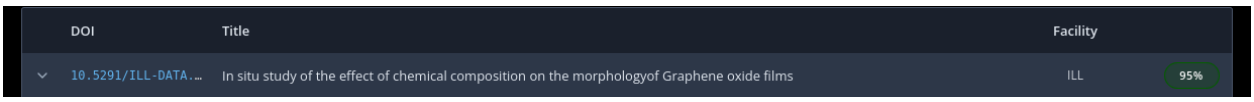


Figure 13. Collapsed result entry

5.2.7.2 Expanded result entry

When the user clicks on the collapsed row and not on the DOI link, the result entry expands and provides more information (Figure 14).

The expanded entry first provides an AI-generated relevance explanation describing why the selected entry is relevant to the user query. Right below, the user can find the result feedback widget. Here the user can tell the system if the result entry has been placed in the correct group (Match, relevant or suggested) or not. The user can also select “not relevant”. We encourage the user to provide as much feedback as they can, as we will use this information to further train the system.
The interface then presents: DOI, title, authors list, and full abstract. The bottom row shows the facility and the instrument where the data has been acquired and, lastly, the publication year, when available, and the date on which the associated data became or are scheduled to become publicly accessible.
One last functionality is “view metadata”, this button provides access to the raw information stored in the PaN-Finder system that has been collected from the data sources. This functionality is targeted for advanced users and application testers to facilitate validation of the result entry relevance. The metadata can also be downloaded for further use.

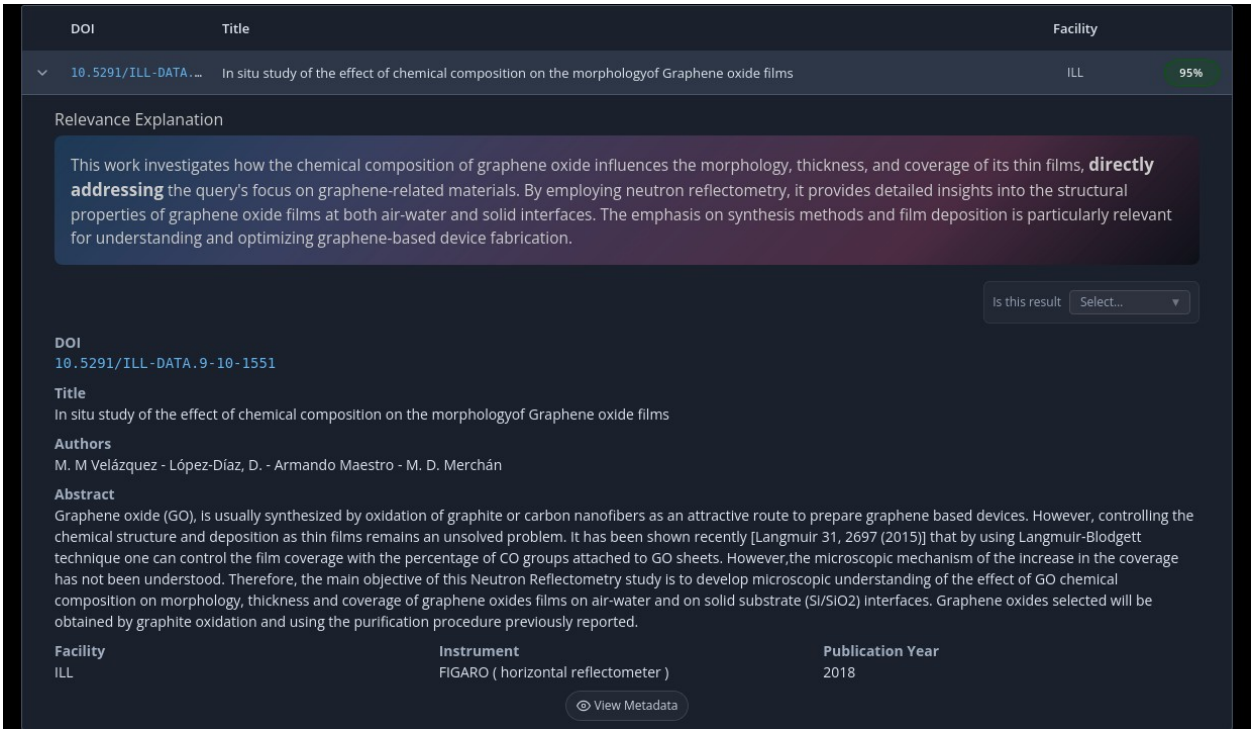


Figure 14. Expanded result entry

6 Technical Overview

Users interested in how PaN-Finder works under the hood and the technical details of its implementation should visit the deliverables github repository, which this document is part of, and the code repository.

- Deliverables: https://github.com/PaN-Finder/deliverables
- Code: https://github.com/PaN-Finder/pan-finder

6.1 System Architecture

PaN-Finder is built with a modern scalable architecture. Figure 15 presents the overview of design and diagram of the application as a whole.

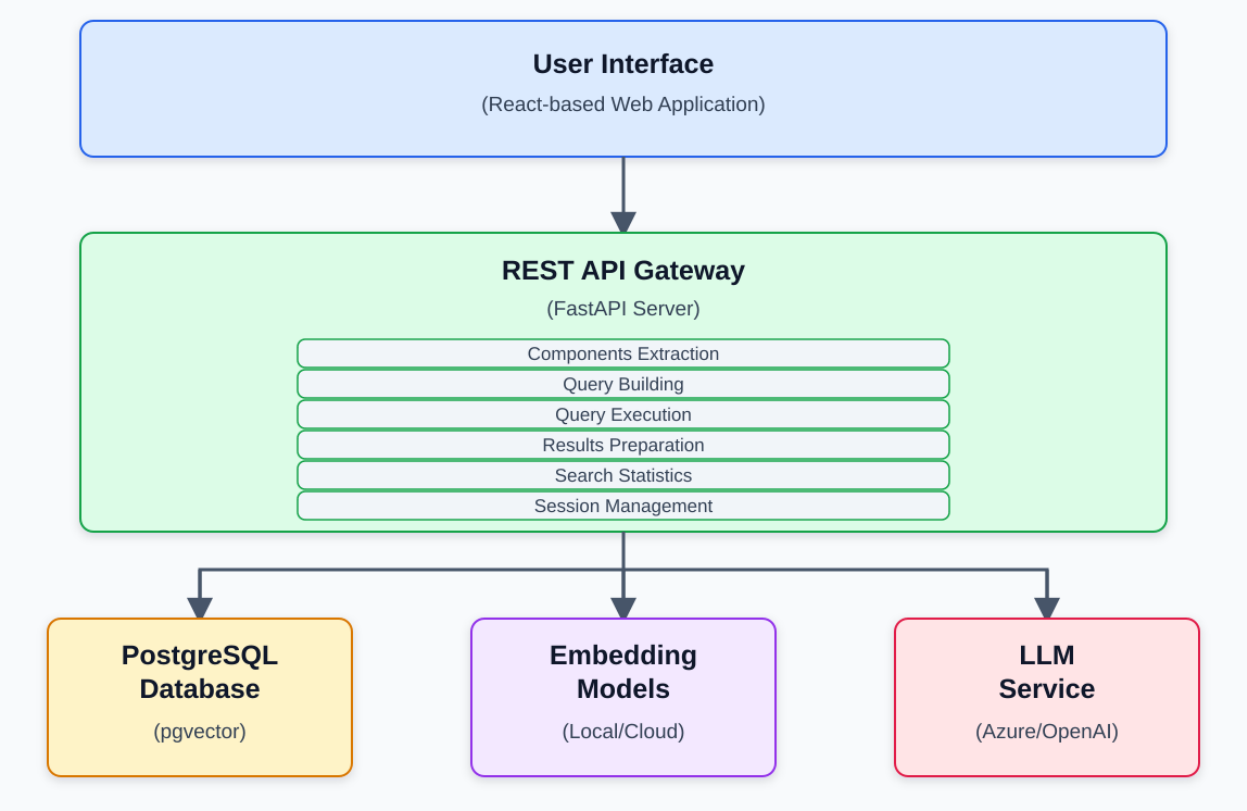


Figure 15. System Architecture

6.1.1 Components

6.1.1.1 Frontend

The frontend is a single-page web application written in TypeScript based on the React framework.

6.1.1.2 Backend

The backend is a REST application written in Python using FastAPI.

6.1.1.3 Database

The database is an instance of PostgreSQL with the pgvector and unit-related extensions.

6.1.1.4 LLM

PaN-Finder is currently configured to use a cloud-based instance of Azure OpenAI through the OpenAI API. It also supports other models, both cloud-based and local. It has been briefly tested with a local Ollama deployment using LiteLLM, although the results with the standard model were inferior to the current configuration.

6.2 Data collection

A series of custom scripts have been created to collect data from each facility. These scripts are currently available in Deliverable 1. They will be transferred in the code repository in the near future as soon as the PaN-Finder application data collection and maintenance strategy is finalized.
Each facility has a related script that leverages the main API to get a complete list of open data, which is then refined and integrated with additional information from multiple sources.
Originally the complete list of open data was obtained through the PaNOSC interface. As the team became more proficient in the technicalities of the data infrastructure for each facility, it has become clear that the PaNOSC interface does not provide enough details about each entry, although it is still useful and used, and the information collected through it needs to be integrated with other sources.

The output of the data collection is one file per facility containing all the information for every entry collected from the facility. These files are in JSON format.

6.3 Data Ingestion and Processing

The data-ingestion process imports the information contained in the JSON files created by the data collection and inserts it in the PaN-Finder database. The ingestion process saves the raw information, and performs additional data processing and curation, preparing the collected data for use by the application search technology.

The data-ingestion process consists of the following pipeline:

- **Summarization**:
  Summarize each entry with a limited number of words (currently 100). An LLM is used for this task and a purpose-built prompt is used.
- **Normalization:**
  Standardize metadata fields, perform unit conversion to SI, and text cleaning
- **Embedding:**
  All relevant text is converted to vector embeddings using the selected model. (Currently we use Sentence Transformers from `all-MiniLM-L12-v2`)
- **Indexing:**
  All entries and all filter keys are indexed to improve the search performance.

6.4 Search Technology

PaN-Finder combines multiple search technologies. They are implemented directly at the database level by creating and submitting a hybrid query. This design is the result of selecting the best solution for performance improvements. The search technologies implemented are:

1. **Vector Similarity Search (pgvector):**
   Search for semantically similar documents using embeddings.

2. **Full-Text Search:**
   Keyword search with PostgreSQL's capabilities

3. **Structured Search:**
   Metadata filters, numerical range queries, unit-aware comparisons

4. **Reciprocal Rank Fusion (RRF):**
   Combines rankings from multiple methods

5. **LLM-Based Query Parsing:**

Understands intent and extracts conditions, parameters and keywords from the user prompt. The application has been developed to convert natural language to structured queries on a best-effort basis.

6.5 Ranking Algorithm

Search in the PaN-Finder leverages the built-in functionalities present in PostgreSQL. It is based on a hybrid approach, implementing 5 different methodologies:

1. *Vector Similarity* applied to the AI-generated summary
2. *Vector Similarity* applied to multiple chunks created from the full entry,
3. *Full-Text* search using keywords
4. Full match of *Structured Filters* created from explicit and implicit conditions included in the prompt.
5. Partial match of the individual *Structured Filters* created from explicit and implicit conditions included in the prompt.

When a user submits a query, the back end executes the following steps:

1. User query is sent to the configured LLM agent for component extraction: intention, keywords, filters.
2. A query is generated by combining the 5 methodologies reported above into a single hybrid SQL query
3. PostgreSQL executes the query leveraging standard SQL and pgvector functionalities
4. PostgreSQL merges scores from different methodologies in a single overall score using weighted RRF. This step is included in the SQL hybrid query.
5. Results are ranked by overall score
6. Results are returned to the user
7. An AI-generated explanation of the relevance is generated when a user selects the detailed view of a result.

For more technical information about the hybrid query and how it is created, please review the document `QUERY_BUILDER.md`
(https://github.com/PaN-Finder/pan-finder/blob/main/docs/QUERY_BUILDER.md)

6.6 Data Model

All the original ingested data and the additional data created during the curation process are stored in the PostgreSQL database. The core tables of the application are:

- *document*
  It contains the raw information, the summary, the title, the abstract, the DOI of the entry and the relevant vectorization.

- *chunk*
  It contains the vector representation of each individual chunk extracted from each entry.
- *filter*
  It contains the custom representation of each pair (key, value) extracted from each entry and further curated. It contains information about units and data type.
- *facility*
  List of the facilities with their abbreviation and full name.
- *filter_description*
  It contains the description of each filter key, including their vector representation used to find similar keys.

An additional table named *feedback* is present. This table is used to save every user query submitted, the query components, the hybrid query, the results set and the additional feedback provided by users.

Some key technical solutions are worth mentioning:

- 384-dimensional vector embeddings are used for similarity searches
- units are automatically converted to SI before being used in searches
- Strong typing is used in filters (boolean, numeric, timestamp, SI units)
- JSONB is used to allow flexibility when storing the original data

7 Examples and Use Cases

7.1 Example 1: Finding Graphene Datasets

7.1.1 Scenario

A materials scientist wants graphene research across European facilities.

7.1.2 Initial Query

graphene

7.1.3 Results

The result set does not contain any Match results, so there are only “relevant” and “suggested” results shown.

7.1.4 Screenshot

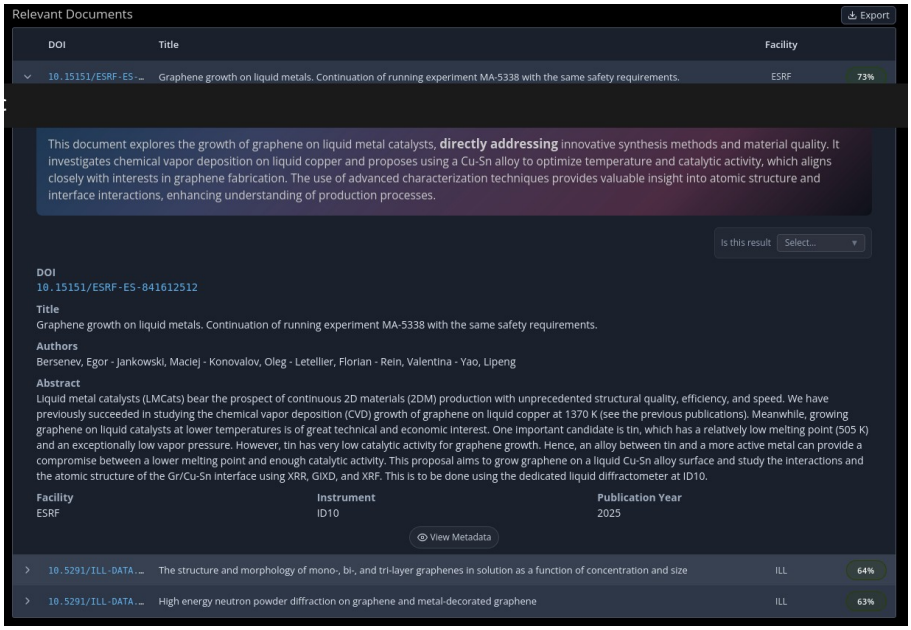


Figure 16. Relevant results for query "graphene"

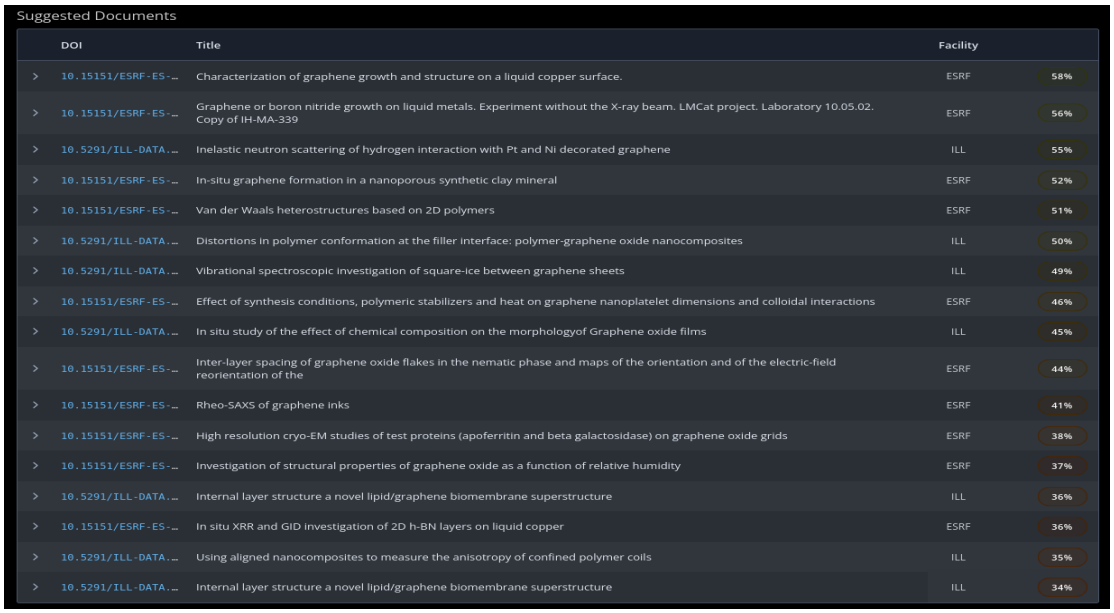


Figure 17. Suggested results for query "graphene"

7.1.5 Follow-up queries

- Find dataset about graphene acquired at ILL
- graphene OR "graphene oxide" OR reduced graphene
- Find graphene datasets where temperature is between 10K and 300K
- Find dataset where sample contains graphene and technique is reflectometry

7.2 Example 2: Temperature-Dependent Experiments

7.2.1 Scenario

A researcher wants phase transition experiments at specific temperatures.

7.2.2 Initial Query

temperature-dependent phase transition

7.2.3 Screenshot

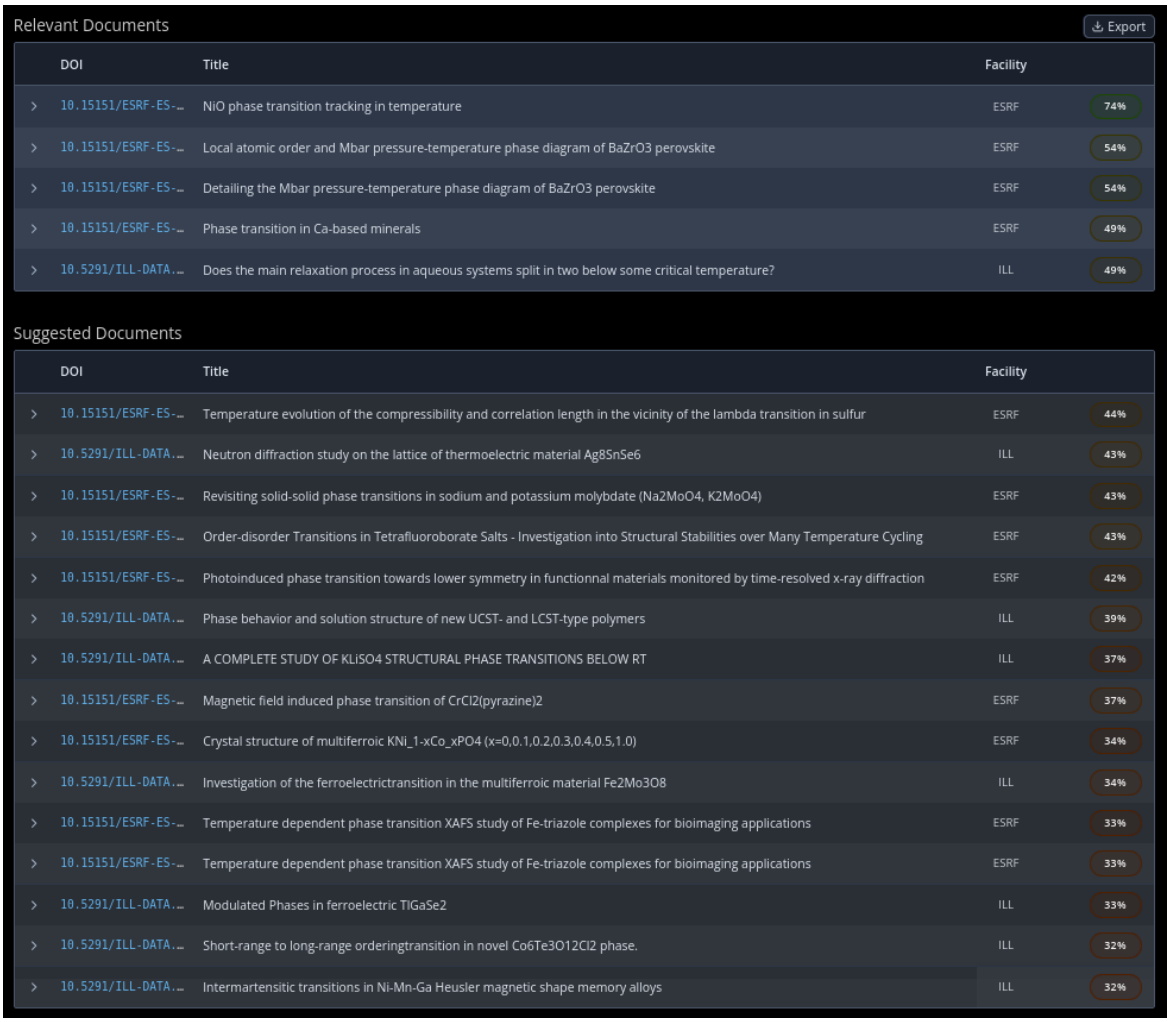


Figure 18: Relevant and suggested document for temperature-dependent phase transition

7.2.4 Follow-up Queries

- Find experiments with temperature between 99K and 101K showing phase transition
- Phase transition in water ice between 270K and 273K
- Water experiments with temperature > 273K and < 293K using neutron scattering

7.3 Example 3: Energy Range Searches

7.3.1 Scenario

A physicist wants experiments with specific energy ranges.

7.3.2 Initial Query

Dataset with energy between 1 keV and 50 keV

7.3.3 Screenshot

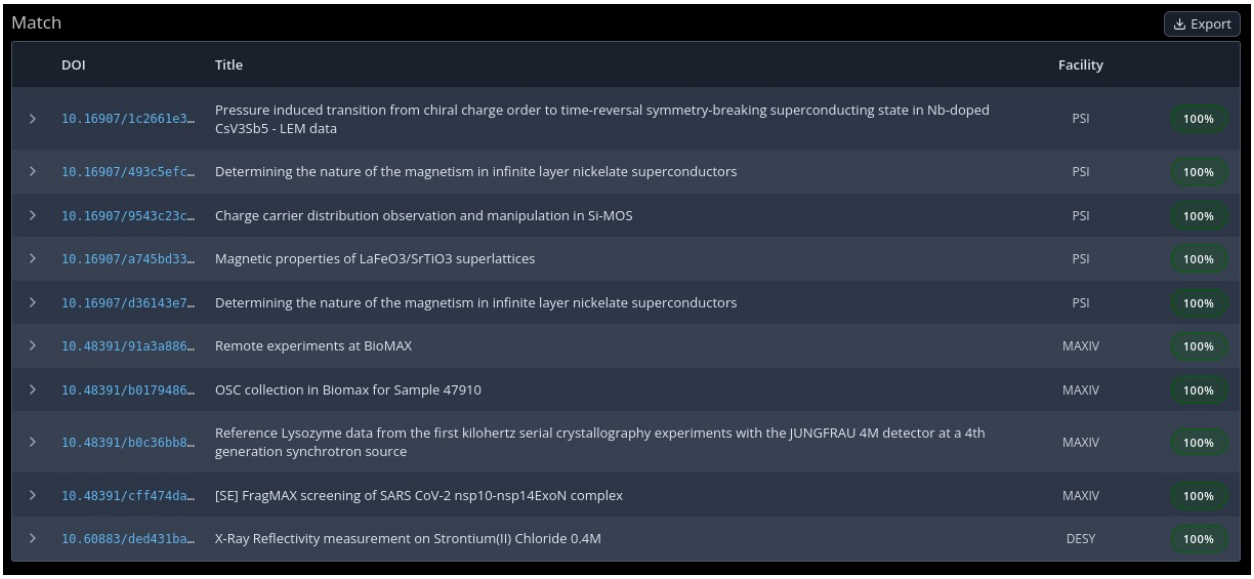


Figure 19. Results with query “Dataset with energy between 1 keV and 50 keV”

7.3.4 Follow-up Queries

- X-ray experiments with energy between 10 keV and 100 keV
- Neutron scattering with energy > 100 meV and < 500 meV
- Aluminum experiments with X-ray energy 80 keV
- Experiments with wavelength 1 Å

7.4 Example 4: Facility-Specific Queries

7.4.1 Scenario

A user wants data from a specific facility or facilities to perform cross-facility comparison.

7.4.2 Initial Query

facility:ESS graphene

7.4.3 Screenshot

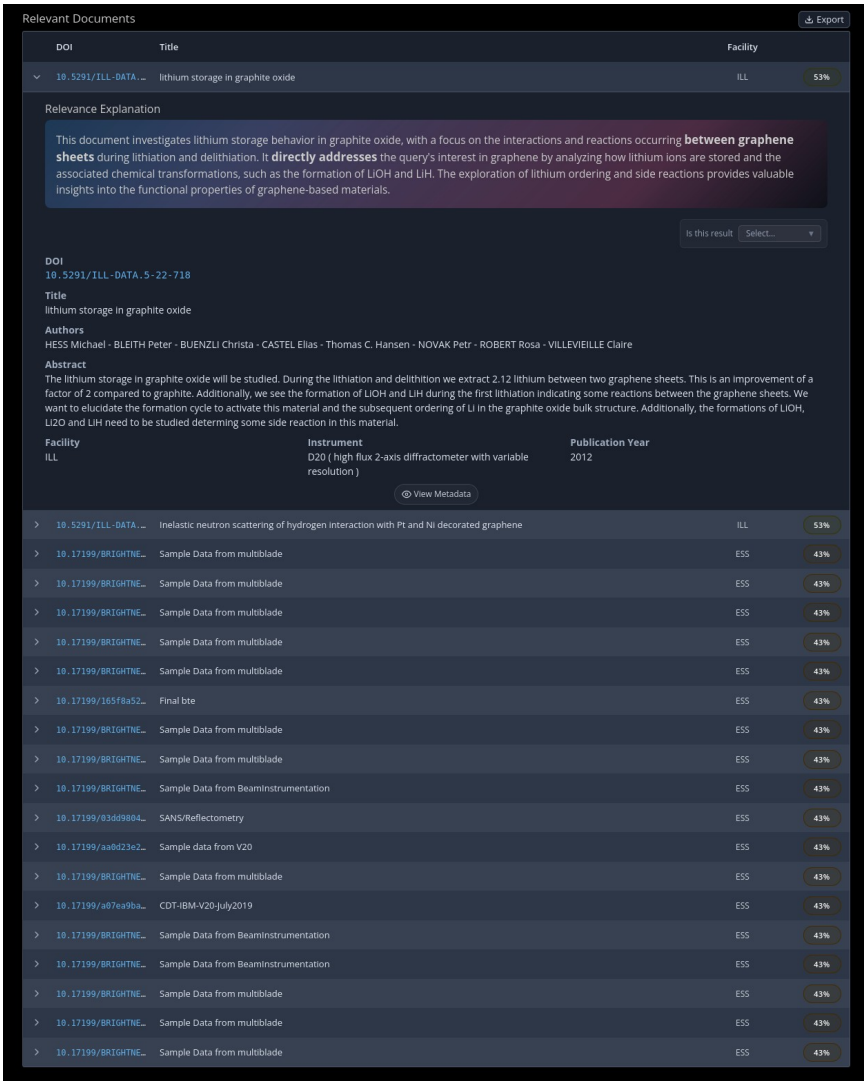


Figure 20. Results for query: facility:ESS graphene

7.4.4 Additional Queries

1. (facility:ESS OR facility:ILL) AND neutron scattering
2. ESRF ID23-1 experiments on proteins
3. ESS DREAM instrument datasets
4. (graphene AND diffraction) AND (facility:ESS OR facility:ESRF)

7.5 Example 5: Combined Parameter Search

7.5.1 Scenario

A researcher needs experiments with multiple specific parameters.

7.5.2 Initial Query

Find datasets about Protein using technique SAXS

7.5.3 Screenshot

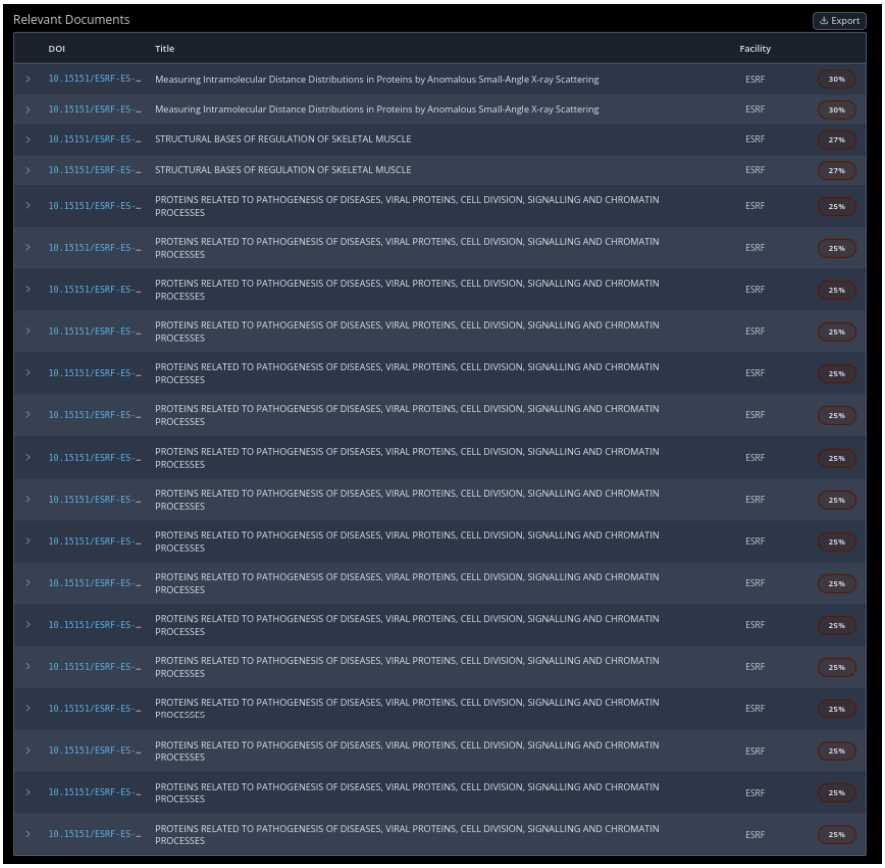


Figure 21. Results for query "Find datasets about protein using the SAXS technique”

7.5.4 Follow-up Queries

- Protein crystallography at 100 K using X-ray diffraction
- Find protein datasets where temperature is between 90 K and 110 K,
  energy is between 12 keV and 14 keV, and facility is ESRF
- Graphene experiments with temperature < 300K but NOT using X-ray
- Lysozyme protein crystallography at 100 K with wavelength 0.98 Å

8 API Documentation

This section is intended for developers and advanced users.

8.1 REST API Endpoints

The current base URL of the publicly available PaN-Finder application is:

`https://pan-finder.panosc.ess.eu/api/`

List of endpoints:

- *Endpoint:* `/search/structured`
  Method: `POST`
  Description: Search using explicit components: intention, keywords and filters

- *Endpoint:* `/document/{doi}`
  Method: `GET`
  Description: Retrieve the document details given its DOI

- *Endpoint:* `/feedback/submit`
  Method: `POST`
  Description: Submit user feedback

- *Endpoint:* `/session/create`
  Method: `POST`
  Description: Create a user session using Turnstile

- *Endpoint:* `/health`
  Method: `GET`
  Description: Health check

8.2 Authentication

No account is required to use PaN-Finder; nevertheless, a session and a human verification is required. To use the API directly, the user is required to perform the following two steps:

1. Create a session with a Turnstile token.
   Execute a POST request to /session/create with the following payload
   `{"turnstile_token":"…"}`.
   This step is required only before the first search
2. Include the following header in each subsequent request:
   `X-Session-ID: UNIQUE_SESSION_ID`
   where UNIQUE_SESSION_ID is the token returned by the previous operation

8.3 Rate Limits

At the moment there are no explicit limits, although the session and human verification act as a rate limit. The team is monitoring the usage and will impose rate limits and throttling if abuse is detected.

8.4 API Usage Examples

The examples below assume that an API session has already been created and that the returned session ID is available. See the Authentication section for details.

8.4.1 Simple Command Line Search

```
curl -X POST https://pan-finder.panosc.ess.eu/api/search \
  -H 'Content-Type: application/json' \
  -H 'X-Session-ID: YOUR_SESSION_ID' \
  -d '{"query": "graphene neutron scattering"}'
```

8.4.2 Command Line Search with Filters

```
curl -X POST https://pan-finder.panosc.ess.eu/api/search \
  -H 'Content-Type: application/json' \
  -H 'X-Session-ID: YOUR_SESSION_ID' \
  -d '{"query": "dataset from temperature experiments performed
```

```
                at ESS or ESRF between 2020-01-01 and 2024-12-31"}'
```

8.4.3 Command Line Search from query components

```
curl -X POST https://pan-finder.panosc.ess.eu/api/search/structured \
  -H 'Content-Type: application/json' \
  -H 'X-Session-ID: YOUR_SESSION_ID' \
  -d '{
       "structured_data": {
        "intention": "graphene",
        "keywords": ["graphene"],
        "filters": {
         "logic": "AND",
         "conditions": [
          {
           "name": "temperature",
           "operator": "BETWEEN",
           "value": [99, 101],
           "unit": "K"
          }
         ]
        }
       }
      }'
```

8.4.4 Python Simple Search

```
import requests
API_BASE = "https://pan-finder.panosc.ess.eu/api"
response = requests.post(
  f"{API_BASE}/search",
json={"query": "graphene"}, headers={"X-Session-ID": "YOUR_SESSION_ID"}
)
print(response.JSON())
```

8.4.5 JavaScript Simple Search

```
const API_BASE = "https://pan-finder.panosc.ess.eu/api";
async function search(query) {
  const response = await fetch(
    `${API_BASE}/search`, {
      method: "POST",
headers: {"Content-Type": "application/json", "X-Session-ID":
"YOUR_SESSION_ID"},
      body: JSON.stringify({ query: query })
    }
  );
  return await response.JSON();
}
```

9 Troubleshooting and FAQ

9.1 Common Issues and Solutions

- *Problem*: No results
  Possible Cause: Query too specific
  Solution: Relax the search conditions, broaden search terms, remove filters
- *Problem*: No results
  Possible Cause: No matching data
  Solution: Change keywords and conditions
- *Problem*: Results with low relevance
  Possible Cause: Query too broad or wrong terms
  Solution: Narrow the search criteria, add additional conditions or use synonyms
- *Problem*: Results with low relevance
  Possible Cause: Query misinterpreted
  Solution: Rephrase the query, use synonyms.

9.2 Frequently Asked Questions

Q: Do I need an account?
A: No, PaN-Finder is publicly accessible without registration. It requires only human verification.

Q: Is PaN-Finder free?
A: Yes, the PaN-Finder is completely free.

Q: Can I download data?
A: PaN-Finder does not have access to the raw data, therefore it does not provide raw data download. PaN-Finder allows the download of the raw metadata related to each entry. PaN-Finder refers the user back to the facility for raw data download.

Q: How often is data updated?
A: As PaN-Finder transitions into maintenance mode, the data update is planned monthly. As soon as the update strategy is finalized, the update cycle will be posted and exposed in the application.

Q: Can I search for data that are still under embargo?
A: PaN-Finder can index a public record whose associated data are still under embargo. However, PaN-Finder does not provide access to restricted data. Access becomes available only when the facility’s embargo period has ended. The record’s data-availability or scheduled-release

information is shown when provided by the source facility.

Q: How do I cite data?
A: PaN-Finder refers users back to the individual facility about citing their data. If data is discovered through the PaN-Finder, please cite the PaN-Finder as indicated in the related section.

Q: Can I use API for bulk downloads?
A: No. The PaN-Finder API is configured to return the most relevant entries found in the database for a given user query. It currently returns a maximum of 20 results per query. This hard limit is set in the backend and can be changed in the backend configuration.

Q: What facilities and repositories are supported?
A: Public data is harvested from the following facilities and repositories: ESS, ESRF, ILL, MAX IV, PSI, DESY, Human Organ Atlas. More will be added.

Q: How does PaN-Finder handle privacy?
A: PaN-Finder does not collect any personal information. The only information collected is the queries submitted together with the results returned and the internal status of the app.

Q: Can I request that a dataset be added?
A: Yes, you can as long as the information about the dataset is associated with one of the supported facilities, it is open, public and has a registered DOI. Contact the PaN-Finder team providing the DOI, the facility and, possibly, a few examples of queries that should return the new entry.

Q: Is there a search limit?
A: At the moment, the PaN-Finder does not impose hard limits for normal use.

10 License Information

The PaN-Finder material and deliverables are all licensed under CC BY-SA 4.0 (Creative Commons Attribution-ShareAlike 4.0), while the software is distributed under GPL v3 (GNU General Public License version 3).
The data forming the PaN-Finder body of knowledge is all public data and we refer users to the data provider (facility) for the license of the specific result entry.

10.1 Content License

CC BY-SA 4.0 (Creative Commons Attribution-ShareAlike 4.0 International)

- **You can:** Share, adapt for any purpose (including commercially)
- **Conditions:** Attribution, ShareAlike
- **Full text:** https://creativecommons.org/licenses/by-sa/4.0/

10.2 Software License

GPL v3 (GNU General Public License version 3)

- **You can:** Use, modify, distribute for any purpose
- **Conditions:** Make source code available, modified versions must be GPL v3, include license notices
- **Full text:** https://www.gnu.org/licenses/gpl-3.0.en.html

10.3 Data License

Each individual datasets are covered by the facility data policy and license. Always check on the facility website and the DOI landing page before reuse.

11 Glossary

11.1 General Terms

API
Application Programming Interface. Protocols for interacting with software

DOI
Digital Object Identifier. Persistent unique identifier for scientific works

Federated Search
Search system querying multiple data sources, presenting unified results

Metadata
Data about data. Descriptive information providing context to raw data

Open Access
Research outputs distributed online, free of access barriers

11.2 Scientific Terms

Angstrom (Å)
Unit of length equal to $10^{\wedge\text{-}10}$ meters, for atomic-scale measurements

Beamline
The path that a particle beam takes from source to experimental station.
Specific experimental setup/device at synchrotrons.

Crystallography
Science of determining atomic arrangement in crystalline solids

Diffraction
The bending or spreading of waves as they pass through an aperture or around an obstacle or around obstacle

eV
electronvolt - energy unit = kinetic energy of electron accelerated by 1 volt

Instrument
Specific experimental setup/device at a neutron research facility

Neutron Scattering
Technique using neutrons to probe material structure and dynamics

Phase Transition
Physical process of transition between states of matter

Synchrotron
Cyclic particle accelerator producing high-intensity X-rays

Wavelength
The spatial period of a wave. Distance over which wave shape repeats

X-ray
Electromagnetic radiation with wavelengths 0.01–10 nanometres

11.3 PaN-Finder Specific Terms

Chunk
Segment of document text used for embedding and vector search

Embedding
Numerical representation of text capturing semantic meaning

Filter
Metadata associated with documents for structured searching

LLM
Large Language Model. AI for natural language understanding/generation

pgvector
PostgreSQL extension for vector similarity search

RRF
Reciprocal Rank Fusion. Method for combining rankings from multiple search systems

SSE
Server-Sent Events - technology for streaming search results in real-time

Turnstile
Cloudflare’s bot protection service

11.4 Facilities and Entities Acronyms

ESS, European Spallation Source
Location: Lund, Sweden
Type: Facility
Source: Neutrons
In commissioning phase

ESRF, European Synchrotron Radiation Facility
Location: Grenoble, France
Type: Facility
Source: X-rays
Operational since 1994, 40+ beamlines

ILL, Institut Laue-Langevin
Location: Grenoble, France
Type: Facility
Source: Neutrons
Operational since 1971, 40+ instruments

MAX IV, MAX IV Laboratory
Location: Lund, Sweden
Type: Facility
Source: X-rays
Operational since 2016, 10+ beamlines

PSI, Paul Scherrer Institute
Location: Villigen, Switzerland
Type: Facility
Source: Synchrotron, spallation neutron source, muon source, and X-ray free-electron laser
Operational since 1988, ~40 instruments

DESY, Deutsches Elektronen-Synchrotron
Location: Hamburg, Germany
Type: Facility
Source: X-rays
Operational since 1965, ~30 beamlines

HOA, Human Organ Atlas
Location: Grenoble, France
Type: Data Repository
Affiliated with ESRF

PaNOSC, Photon and Neutron Open Science Cloud
Location: Europe
Type: Horizon 2020 project

OSCARS, Open Science Cloud for Accelerator-based Research Services
Location: Europe
Type: Horizon Europe project

12 References and Further Reading

12.1 Project References

1. **PaN-Finder Application:** https://pan-finder.panosc.ess.eu/
2. **OSCARS Project:** https://oscars-project.eu/
3. **PaN-Finder OSCARS Page:** https://oscars-project.eu/projects/pan-finder-photon-and-neutron-federated-knowledge-finder
4. **PaNOSC:** https://www.panosc.eu/
5. **EOSC Portal:** https://eosc-portal.eu/

12.2 Technical References

1. **PostgreSQL:** https://www.postgresql.org/
2. **pgvector:** https://github.com/pgvector/pgvector
3. **FastAPI:** https://fastapi.tiangolo.com/
4. **React:** https://react.dev/
5. **Sentence Transformers:** https://www.sbert.net/
6. **Ollama:** https://ollama.com/
7. **LiteLLM:** https://github.com/BerriAI/litellm

12.3 Project Repositories

- **Code Repository:** https://github.com/PaN-Finder/pan-finder
- **Deliverables:** https://github.com/PaN-Finder/deliverables

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All final decisions, interpretations, and responsibility for the work in this deliverable remain with the author(s).